Jiyanshu Singh

B.Tech Biotechnology student at NIT Rourkela | Building ML systems

Contact

■ jiyanshusingh1@gmail.com | ■ +91 8218608077 | ■ github.com/jiyanshusingh | ■ linkedin.com/in/jiyanshusingh

Education

B.Tech Biotechnology · National Institute of Technology, Rourkela

Expected 2027 · CGPA: 7.09/10 · Relevant coursework: Data Structures & Algorithms, Statistics, Mathematics for Engineers

Experience

Freelance Data Scientist / ML Engineer *Dec 2023 – Present*

End-to-end delivery: Built and deployed 4 production web applications (A/B Testing Dashboard, Sales CRM, Invoice Generator, Hydrogel Portal) — all with Dockerized FastAPI/Flask backends on Render/GitHub.

Statistical consulting: Delivered A/B test analysis (p-values, Bayesian Monte Carlo, power analysis) for small clients.

Projects

AI Trading Engine [Code]

Problem: A naive strategy lost ■225k; an XGBoost ML filter that labels 38,500+ trades as net-profitable-after-costs turned it into +■93k profit.

Key results (OOS-validated): 95% reduction in false signals; walk-forward validated across 3 market regimes (bear → consolidation → rally).

Tech stack: Python, XGBoost, FastAPI, Docker, GCP (Compute Engine + Cloud SQL), PostgreSQL, walk-forward backtesting framework.

A/B Testing Dashboard [Code]

Purpose: Multi-page statistical analysis app that analyzes experiment data and recommends Launch / Iterate / Kill using p-values, Bayesian Monte Carlo, power analysis, and sample size calculations.

Impact: Generates consulting-grade PDF reports for stakeholders — deployed at https://ab-testing-dashboard.onrender.com.

Tech stack: FastAPI, Bootstrap 5, SciPy, Docker, Render.

Peptide Hydrogel Prediction [Code]

Purpose: Machine-learning models predicting hydrogel formation of short peptides from amino-acid sequence and terminal modifications.

Key results: ROC-AUC 0.888 (amyloid classification), 0.720 (hydrogel); SHAP-based interpretability; strict leakage prevention on 364-sample benchmark; external-transfer validation on 14 literature entries.

Web portal: Multi-page Flask app with researcher auth, dataset browsing, live training jobs, and de-novo peptide design scoring.

Tech stack: Python, RandomForest/XGBoost/ESM2, SHAP, Flask, Docker, Render.

Sales CRM Dashboard [Code]

Purpose: Full-stack CRM for managing sales leads with KPI dashboard, funnel analysis, revenue charts, CSV export, and basic auth.

Tech stack: FastAPI, Bootstrap 5, Plotly.js, SQLAlchemy, Plotly Dash, Docker, Render.

Invoice Generator [Code]

Purpose: SaaS-ready invoice generation service with PDF export, SQLAlchemy backend, template-based rendering, and Docker deployment.

Tech stack: FastAPI, SQLAlchemy, ReportLab (PDF), Docker, Render.

Skills

Languages: Python (expert), SQL

ML/AI: scikit-learn, XGBoost, RandomForest, protein language models (ESM2), SHAP, feature engineering

Biotech tools: Biopython, peptide representation, A/B testing for bioassays, statistical analysis

Deployment: Flask, Docker, Render, data visualization

Achievements

- Deployed 5 production web applications (FastAPI/Flask + Docker + Render/GCP).
- Labeled 38,500+ trades for ML filter; 95% false-signal reduction; walk-forward OOS validated across 3 market windows.
- 350-sample peptide benchmark with leakage prevention; amyloid ROC-AUC 0.888.

Skills I'm learning: XGBoost, walk-forward testing, Bayesian A/B testing, biotech ML.